

FINANCIAL ANALYSIS REPORT

Stock Price vs Fundamental Variables

Date: 2026-02-12

Stock Price vs Fundamentals Analysis

1. EXECUTIVE SUMMARY

This analysis examines the relationship between stock prices and fundamental variables for Indian IT companies (TCS and Infosys).

KEY FINDINGS:

1. TOP 3 MOST SIGNIFICANT VARIABLES:

- (1) INFY - PAT Growth (p-value = 0.0021)
- (2) TCS - Sales Growth (p-value = 0.0029)
- (3) TCS - PAT Growth (p-value = 0.0191)

2. MODEL PERFORMANCE:

- Infosys (INFY): R-squared = 0.482
- TCS: R-squared = 0.418

3. KEY CORRELATIONS:

- INFY: PAT Growth vs Stock Price = -0.661
- TCS: Sales Growth vs Stock Price = -0.532

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2. TOP 3 MOST SIGNIFICANT VARIABLES

Rank	Company	Variable	Coefficient	P-Value	Significance

1	INFY	PAT_Growth	-749.6342	0.0021	Highly Significant
2	WIPRO	EBITDA_Growth	-26.1678	0.0029	Highly Significant
3	TCS	Sales_Growth	-1825.5290	0.0191	Significant

Significance Levels:

- $p < 0.01$: Highly Significant
- $p < 0.05$: Significant
- $p < 0.10$: Weakly Significant

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3. CORRELATION ANALYSIS RESULTS

Correlation with Stock Price by Company:

Company	Sales Gr	EBITDA Gr	PAT Gr	EBITDA Mar	PAT Mar
TCS	-0.532	-0.313	-0.451	-0.231	0.031
INFY	-0.383	-0.281	-0.661	0.121	-0.267
WIPRO	-0.241	-0.644	-0.430	-0.154	-0.105
TECHM	-0.301	-0.364	-0.423	0.182	0.247

Key Observations:

- INFY: Strong negative correlation with PAT Growth (-0.661)
- TCS: Strong negative correlation with Sales Growth (-0.532)

4. REGRESSION ANALYSIS RESULTS

Model Summary (R-squared):

Company	R-squared	Adj R-squared
TCS	0.4184	0.1947
INFY	0.4823	0.2831
WIPRO	0.4683	0.2637
TECHM	0.2724	-0.0075

Interpretation:

- INFY model explains 48.2% of stock price variation
- TCS model explains 41.8% of stock price variation

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5. REGRESSION COEFFICIENTS

Company	Sales Gr	EBITDA Gr	PAT Gr	EBITDA Mar	PAT Mar
TCS	-1825.5	1727.2	457.2	-1361.6	-956.5
INFY	130.8	331.7	-749.6	-324.9	282.1
WIPRO	19.2	-26.2	-43.2	-37.4	32.6
TECHM	-479.4	111.6	-168.6	-751.5	287.2

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6. STATISTICAL SIGNIFICANCE (P-VALUES)

Company	Sales Gr	EBITDA Gr	PAT Gr	EBITDA Mar	PAT Mar
TCS	0.0191	0.1925	0.0524	0.3406	0.8992
INFY	0.1058	0.2434	0.0021	0.6230	0.2686
WIPRO	0.3197	0.0029	0.0660	0.5294	0.6688
TECHM	0.2101	0.1253	0.0710	0.4549	0.3075

7. CONCLUSIONS AND RECOMMENDATIONS

PRIMARY FINDINGS:

1. PAT Growth is the strongest predictor of stock prices, particularly for Infosys ($p = 0.0021$).
Recommendation: Investors should prioritize companies with strong profit growth.

COMPANY-SPECIFIC INSIGHTS:

2. INFOSYS (INFY):
 - Key Driver: PAT Growth
 - Coefficient: -749.63
 - Model Fit: $R^2 = 0.482$
 - Recommendation: Monitor quarterly PAT growth closely
3. TATA CONSULTANCY SERVICES (TCS):
 - Key Driver: Sales Growth
 - Coefficient: -26.17
 - Model Fit: $R^2 = 0.418$
 - Recommendation: Track sales growth trajectory

INVESTMENT IMPLICATIONS:

4. Use company-specific models for better predictions
5. INFY: Focus on profitability metrics
6. TCS: Focus on revenue growth metrics

LIMITATIONS:

- Sample size: 19 years of overlapping data
- Macro-economic factors not included
- Consider adding quarterly data for future analysis

8. VISUALIZATIONS GENERATED

The following files are available in the output folder:

1. correlation_heatmap.png - Overall correlation matrix
2. company_correlations.png - Company-wise correlations
3. regression_coefficients.png - Regression coefficients
4. r_squared_comparison.png - Model fit comparison
5. pvalue_heatmap.png - Statistical significance heatmap

CSV Data Files:

- overall_correlation_matrix.csv
- company_correlations.csv
- regression_summary.csv
- coefficients.csv
- pvalues.csv
- top_3_significant.csv